# C9555: A Practical Guide to Simulations

## Week 5

**Denys Biriukov – Oct. 15th 2026**

# Get ready for practical session

- Open new terminal `Ctrl + Alt + T`
- Course materials at: https://github.com/VachaLab/C9555
- Any technical issues/problems/questions? Checklist:
  - Access to Wolf.
  - Proper infinity and uv environments.

# Acknowledgments

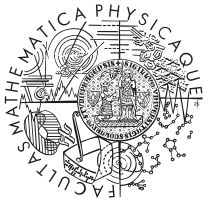
This part of the course is inspired and partially based on:

- Teaching materials of “Spring School in Computational Chemistry 2023” taken place in Finland and prepared by Dr. Hector Martinez-Seara and Dr. Matti Javanainen.
- Teaching course about molecular dynamics simulations at Charles University by prof. Lukasz Cwiklik.









FACULTY
OF MATHEMATICS
AND PHYSICS
Charles University

# More a discussion than monologue

- If you have a quick question to catch up, raise your hand and ask – we will try to fix it and bring you back
- If quick answer does not help, let's chat during a break or after
- This course is mainly practical – theory will be trying to catch up, not vice versa
- Some topics will be repeated – it's fine, you'll cover the same material with multiple teachers

# Get ready for practical session

- `git clone --depth 1 https://github.com/VachaLab/C9555.git C9555_git`
- `cd C9555_git/`
- `cd week05/`

- `bash ../infinity/infinity-temp.sh`
- `bash ../uv/uv-activate.sh`
- `bash ../uv/jupyter-start.sh`

# What is today?

- Week 5:
- How to run MD simulations?
- How and why MD is useful
  - Physics behind MD simulations
- GROMACS
  - Perhaps, the fastest freely-available MD engine
- GROMACS in combination with Jupyter notebooks and Matplotlib

| TEACHING WEEK | DATE | LECTURER | TOPIC |
|---|---|---|---|
| 01 | 17.09 | AT | |
| 02 | 24.09 | AT | |
| 03 | 01.10 | IK | |
| 04 | 08.10 | LS | |
| 05 | 15.10 | DB | How to run MD simulations? |
| 06 | 22.10 | DB | |
| 07 | 29.10 | IK | |
| 08 | 05.11 | IK | |
| 09 | 12.11 | | **BREAK** |
| 10 | 19.11 | DB | |
| 11 | 26.11 | DB | |
| 12 | 03.12 | AT/LS | |
| 13 | 10.12 | AT/LS | |

# Today program

Lecture

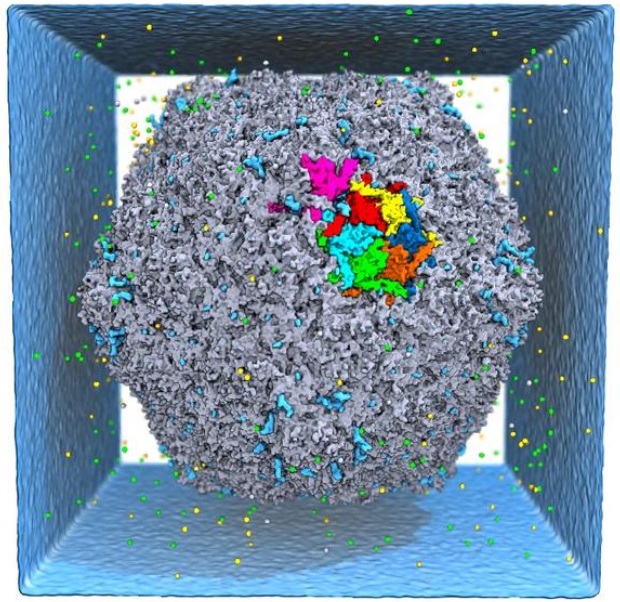
- Recall of basics of MD: why we need it and what physics is behind.
- MD ensembles, relation to thermodynamics.

Practical session

- GROMACS – what is it and how to use it?
- Basic tools to build a system and run a simulation in GROMACS.

# Molecular Simulations: Physics, not Cartoon

- Why do we actually use simulations?
  - Visualizing experiments? Reproducing? Confirming?
- A molecular simulation is a well-controlled *in silico* experiment itself!
  - “Computational microscope”, “computational experiment”.
- Good simulation model – predictive power of simulations.
  - Simulations can initiate new experimental work.
  - Certain computational screenings of novel drugs are more efficient (and much cheaper) than many (if not all) experiments.
- Good model = fundamental physics behind it (approximation of fundamental physical laws).



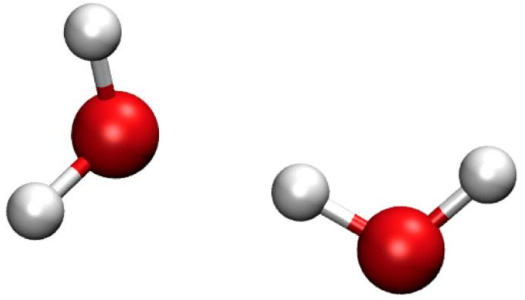
atomistic model of bacteriophage HK97

Coshic et al. Nature, 627, 905–914 (2024)

# Quantum physics?

- Biosimulations usually do not have electrons. Is it legal?
  - Atoms are represented as billiard balls.
  - Averaged effects of electrons are considered through a set of parameters (force fields).
  - Force fields define the potential energy of a system as a function of atomic positions.

- This approximation does not work:
  - Electron transfer processes.
  - Reactive systems.
  - Strong quantum effects.



- Use instead: ab initio molecular dynamics (AIMD), quantum mechanics/molecular mechanics (QM/MM) methods.

# Born-Oppenheimer approximation





# Born-Oppenheimer approximation





# Born-Oppenheimer approximation

Born-Oppenheimer approximation:

- $M_e << M_{nucl}$ ⇒ nuclei move in an averaged field of electrons ⇒ motion of electrons can be separated from motion of nuclei

$$\Psi_{tot} = \psi_{el}(r;R)\psi_n(R)$$

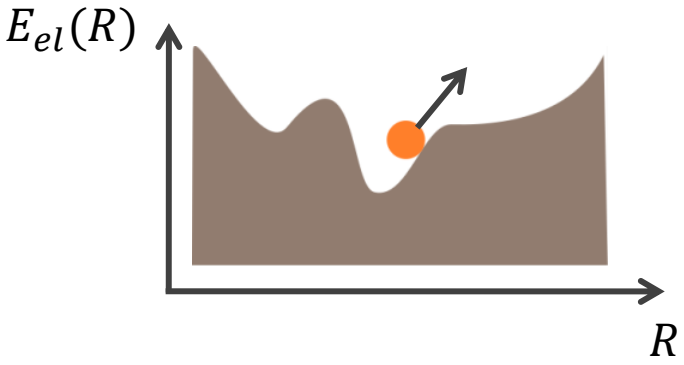
for electrons:

$$H_{el}(r;R)\psi_{el}(r;R) = E_{el}(R)\,\psi_{el}(r;R)$$

for nuclei:

$$[T_n + E_{el}(R)]\psi_n(R) = \mathrm{E}\psi_n(R)$$

- Nuclei move on a potential energy surface $E_{el}(R)$, which can be also denoted as $V(\vec{r})$
- System remains in one electronic state.
- Fails: degenerate energy levels.




- Classical mechanics can be used for nuclei
  - The de Broglie wavelength of nuclei is typically much smaller than the distances between atoms, so their quantum wave effects can often be neglected.
- Not the case of an electron

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  - **Used in biosimulations!**
- … or can be calculated on-the-fly from the electronic wave function in *ab initio* molecular dynamics simulations.
  - Not used in biosimulations, only small systems possible, computationally very expensive, short simulation times, …

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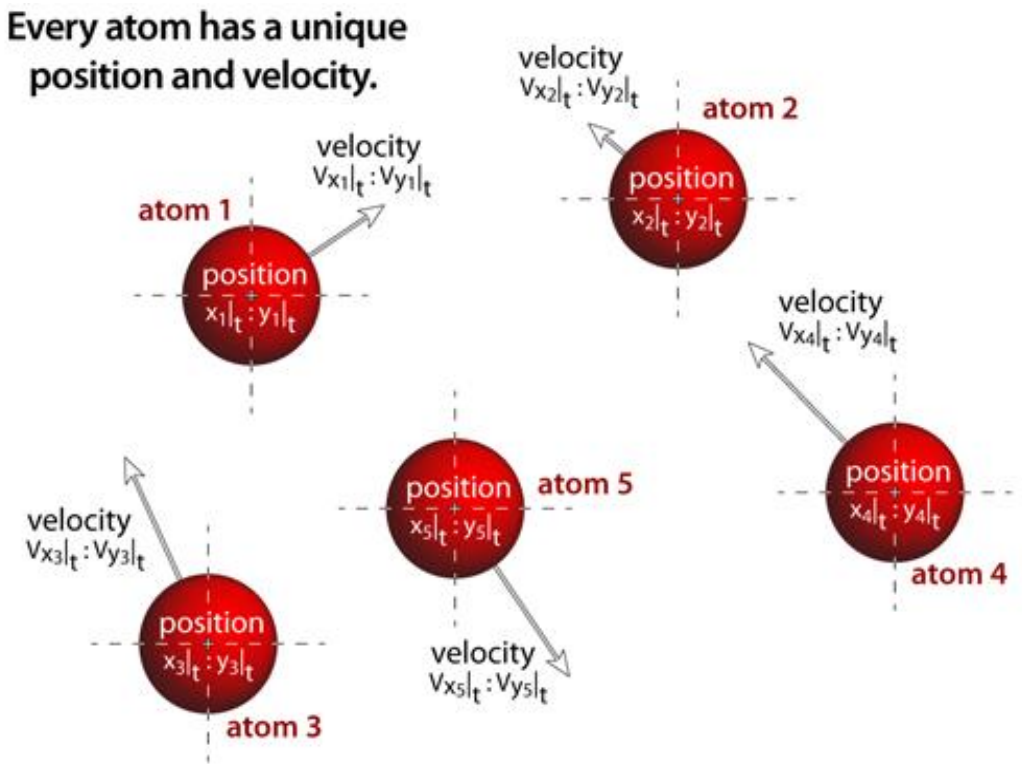
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- current trend − machine-learning potentials (force fields) trained on *ab initio* molecular dynamics
  - not trivial, usually slower than conventional force fields, require high-quality training data, can fail on unfamiliar configurations, system-specific, etc.; available in newest GROMACS versions

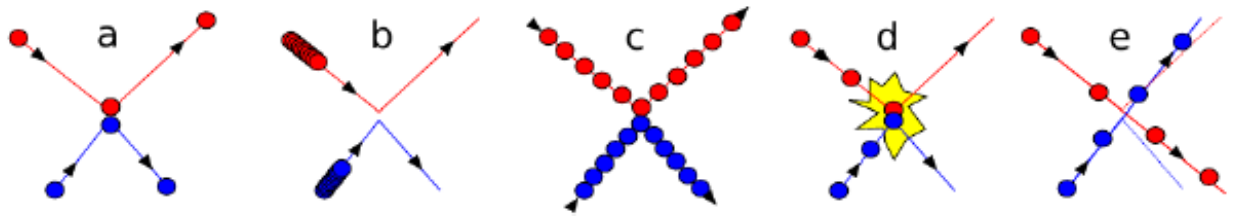
# Time evolution




http://atomsinmotion.com/book/chapter5/md

- Every atom has a unique position (in this example, $x$ and $y$) and velocity ($v_x$ and $v_y$) .
- They move according to Newton's laws of motion.
- Acceleration ($\vec{a}$) acts on every atom and is determined by the force ($\vec{F}$) acting on the atom divided by the atom's mass ($m$).

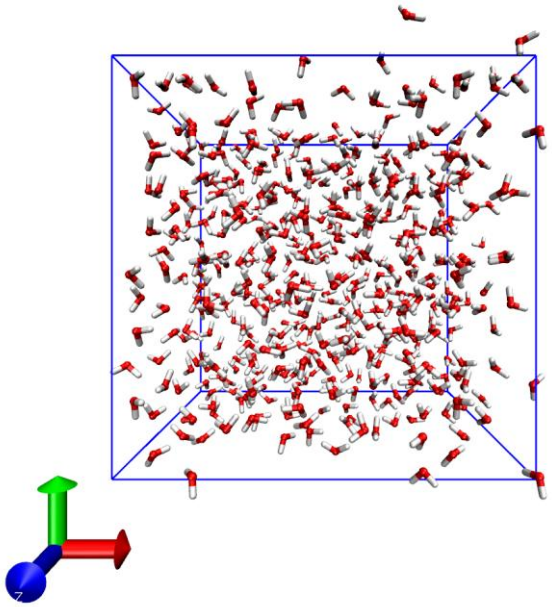
# Time step: optimal Δt

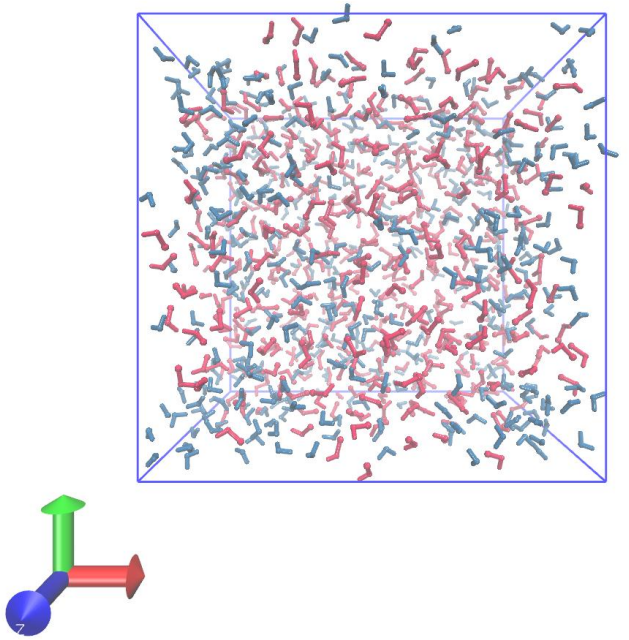



- Time step: not too big, not too small.
- Depends on the force field and its implementation.
- Rule of thumb: at least, one order of magnitude shorter than a characteristic time of the fastest motion in the system.
  - vibration of bonds, time scale ~$10^{-14}$ s, *i.e.*, time step ~1–5 fs

# Micro and macro

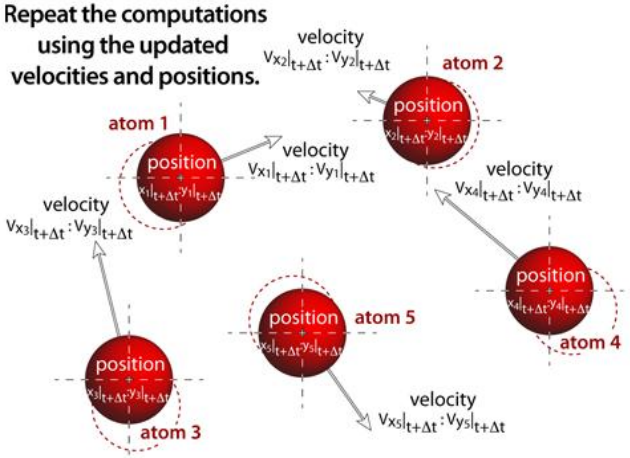
- Newton's equations of motion govern microscopic evolution (atomic positions, atomic velocities).
- They do not explicitly give macroscopic properties (e.g., temperature, pressure, etc.)
- Macroscopic properties are obtained from trajectories via statistical mechanics (statistical thermodynamics).



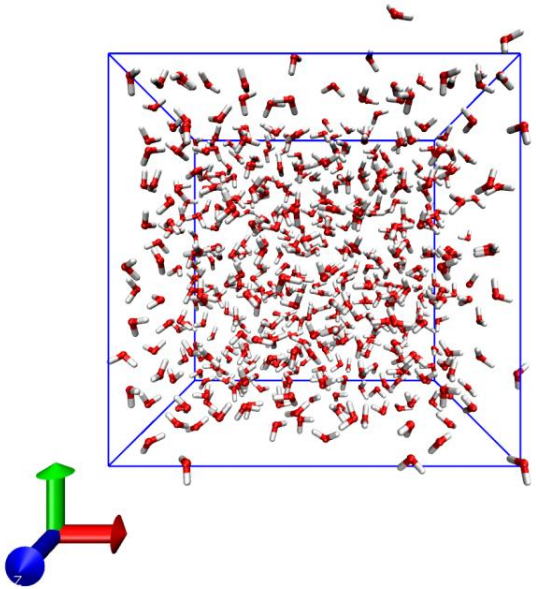


# Micro and macro

- Microscopic properties: $r\ (x, y, z)$, $v\ (v_x, v_y, v_z)$




- Macroscopic properties: $T, P, V, \rho, \eta, \ldots$



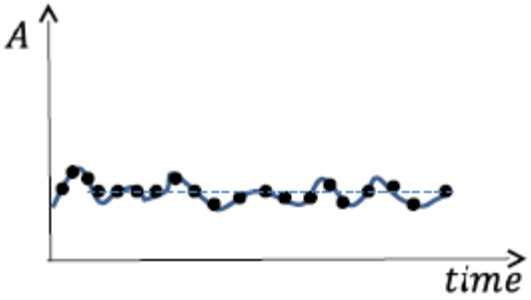


# Ensembles and time

- Ergodic hypothesis: all accessible microstates are equiprobable over a long period of time.
  - Average over ensemble – average over time.

- Macroscopic properties are given by probability distribution within an ensemble.

- In molecular dynamics simulations, we “move” within the ensemble and reproduce its probability distribution.
  - Macroscopic properties can be estimated.
  - Applies to equilibrium molecular dynamics.

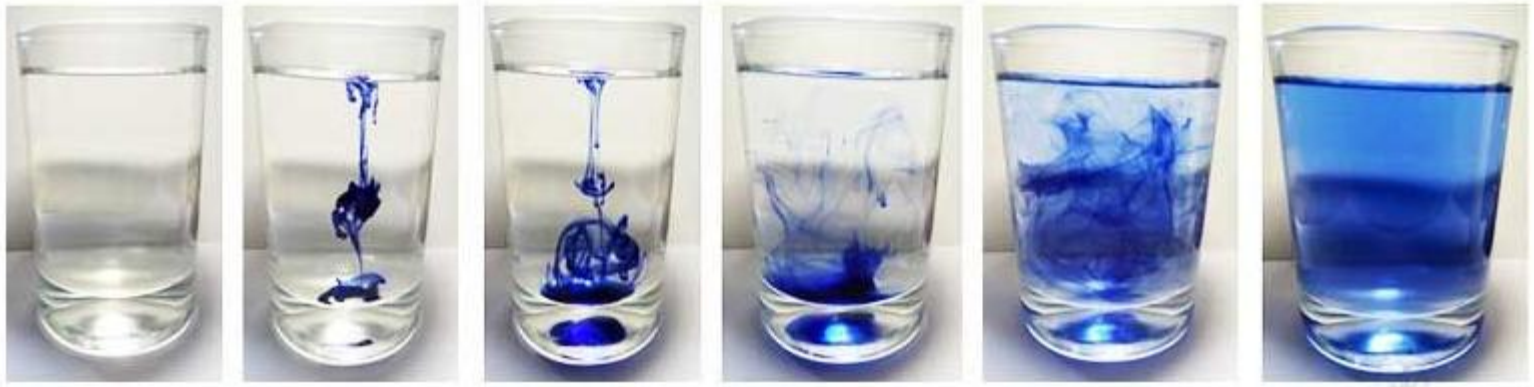



$$\langle A \rangle_{time} = \frac{1}{N} \sum_{i=1}^{N} A(t_i)$$

**under equilibrium**

# Equilibrium vs. non-equilibrium

from https://www.easybiologyclass.com

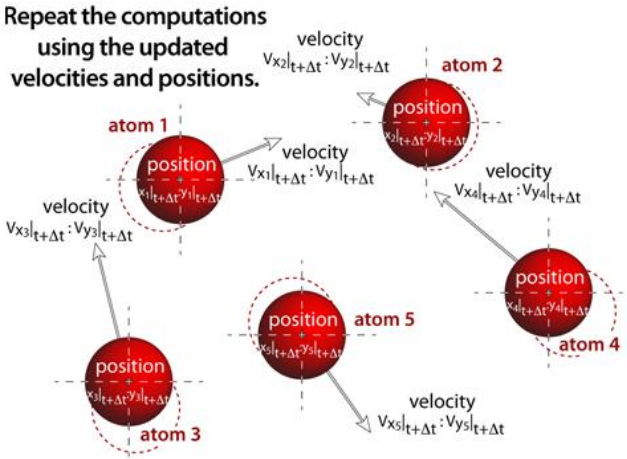


a molecular simulation of ink dissolving in water
is a system out of equilibrium

a molecular simulation of this liquid
is a system under equilibrium

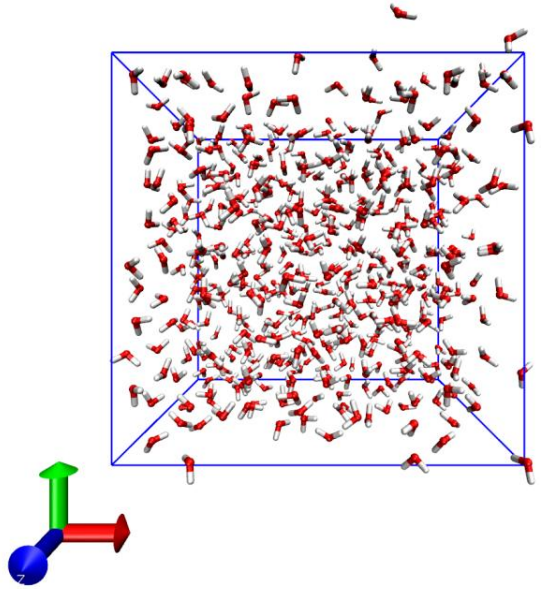
# Micro and macro

- If we have velocities, we can calculate kinetic energy and further, e.g., temperature and pressure.



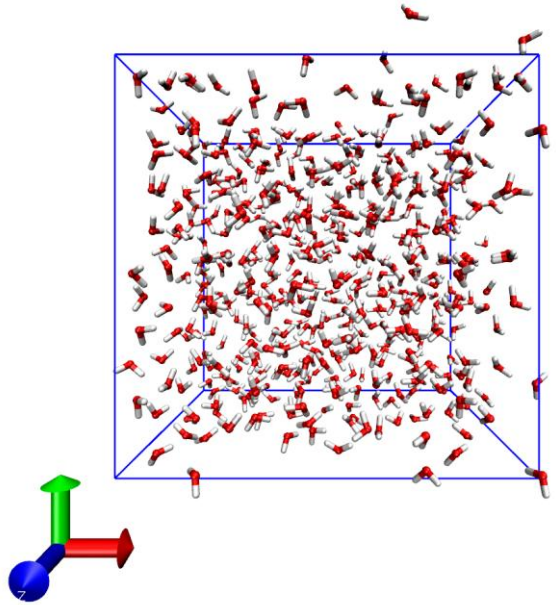

http://atomsinmotion.com/book/chapter5/md

- Relation to pressure:
  - $p = \frac{2N}{3V}\left\langle \frac{1}{2}mv^2 \right\rangle$
- Relation to temperature:
  - $T = \frac{2}{3k_b}\left\langle \frac{1}{2}mv^2 \right\rangle$, $k_b$ – Boltzmann constant



- However, Newton's equations of motion *per se* correspond to the constant energy in the system.
  - Temperature and pressure do not change unless they are adjusted by special algorithms .

# Statistical ensembles in MD



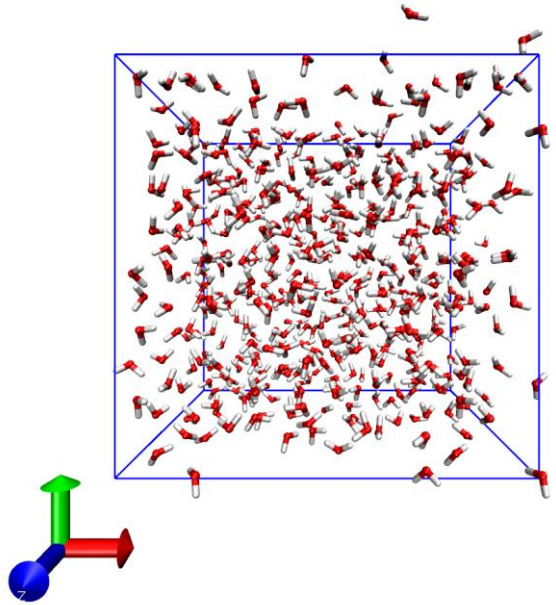
- Typically named by three letters.

- **NVE (microcanonical ensemble):**
  - N – number of particles (constant in a simulation).
  - V – volume (constant in a simulation).
  - E – energy (constant in a simulation).

- Energy is not added nor removed, fixed volume, particles just move in a locked box.
  - Newton's equations correspond to NVE (no external work, constant total energy, no heat exchange).

- Not used in practice: biological simulations are not isolated, big temperature fluctuations, not flexible, convergence and stability issues, etc.

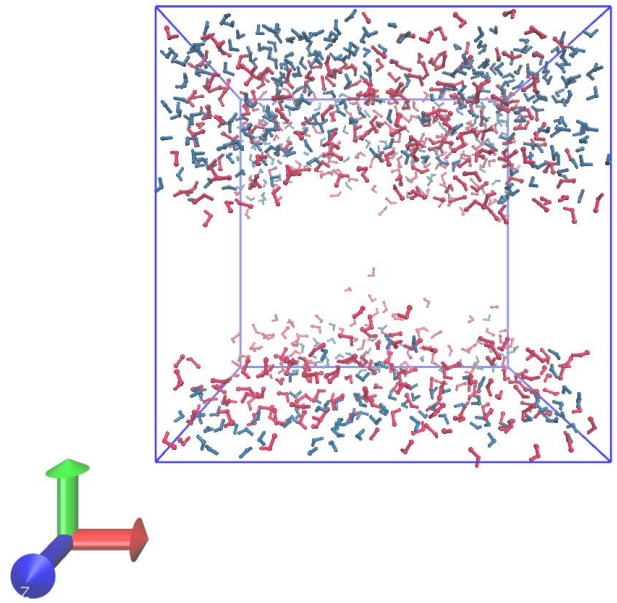
# Statistical ensembles in MD



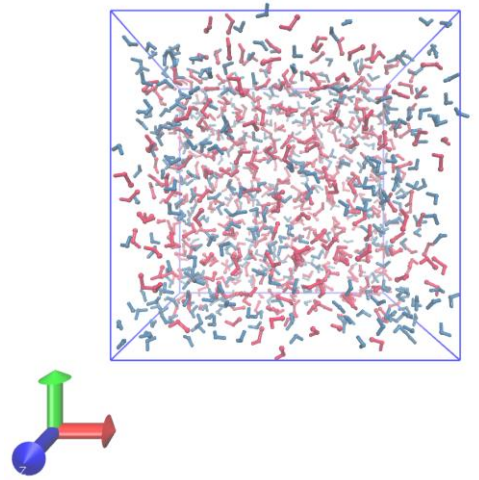
- **NVT (canonical ensemble, isothermal, Helmholtz canonical):**
  - N – number of particles (constant in a simulation).
  - V – volume (constant in a simulation).
  - T – temperature (constant in a simulation).
- Temperature is maintained at a desired value through the exchange of energy with the surroundings (conceptually, of course).
- Sometimes used in bio: simple, stable, realistic enough, excellent for first equilibration stages.
- Disadvantages – volume needs to change in many biosystems, compressibility pressure fluctuations, etc.

# Statistical ensembles in MD

- **NPT (Gibbs canonical ensemble, isobaric-isothermal):**
  - N – number of particles (constant in a simulation).
  - P – pressure (constant in a simulation).
  - T – temperature (constant in a simulation).



- Both temperature and pressure are controlled; pressure is controlled through changing box dimensions.

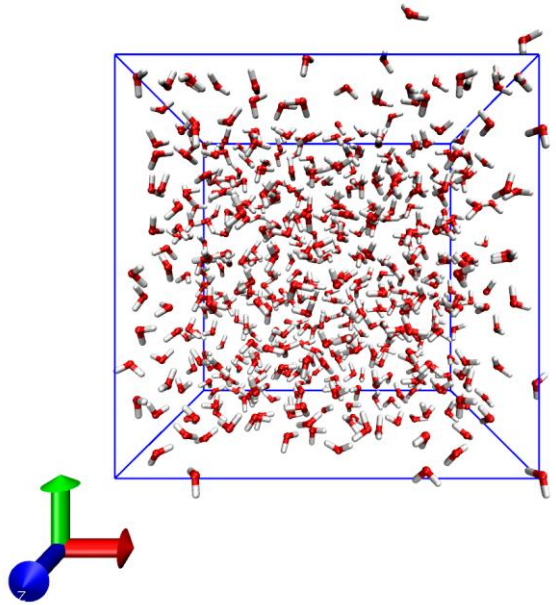


- 99% use-case in biosimulations, replicates biological conditions, stable, flexible, direct comparison to experimental conditions.

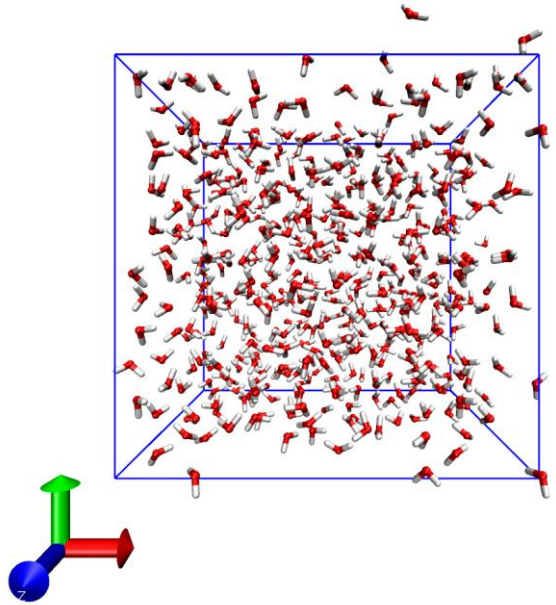
- Disadvantages – basically, technical and computational.

# Statistical ensembles in MD

- Other examples:
  - NPH (isoenthalpic-isobaric: number of particles, pressure, enthalpy).
  - NPE (isobaric: number of particles, pressure, energy).
  - μVT (Grand canonical: chemical potential, volume, temperature).
  - μPT (chemical potential, pressure, temperature).
    - μVT and μPT are usually used in Monte-Carlo simulations.
  - NPγT (number of particles, pressure, surface tension, temperature).
  - …
- I think you will never use them.



# Bonus: energy minimization



- Used to relax the initial structure, removing steric clashes and unrealistic geometries
- Brings the system to a local energy minimum
- Minimizes possible numerical crashes
- There is no reason why you should not use it
- Today we will test energy minimization, NVT, and NPT

Break!

Get ready for practical session!